

Week 2: Theory and In-Class Problems

Linear Algebra I · Fall 2026

Part I: theory and worked examples. Part II: 14 in-class problems with solutions.

Suggested timing: 80 minutes for theory and discussion, then 90 minutes for practice. Use the final two challenges as extension problems if needed.

All vectors use column notation. The notebook retains executable Python for the figures; the PDF omits the code.

Week 1 in one page

Vectors are columns. Linear combinations have the form

$$c_1 \mathbf{v}_1 + \cdots + c_n \mathbf{v}_n.$$

Their span contains every vector produced by such combinations. Independence means that $c_1 \mathbf{v}_1 + \cdots + c_n \mathbf{v}_n = \mathbf{0}$ only for $c_1 = \cdots = c_n = 0$.

For $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$,

$$A\mathbf{x} = x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n.$$

Dot product and angle

Write two vectors of the same dimension as columns:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

The dot product multiplies corresponding components and adds the results. Its output is a **scalar**:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i.$$

For nonzero vectors the angle lies between zero and 180 degrees and satisfies

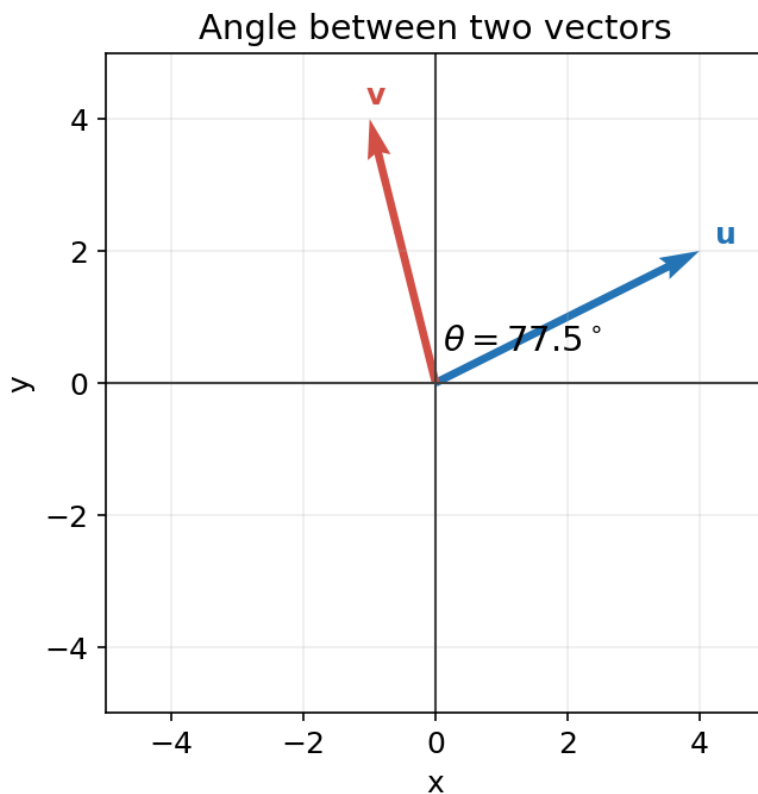
$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

A positive dot product means an acute angle, zero means a right angle, and a negative value means an obtuse angle.

The plotted example:

$$\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \quad \mathbf{u} \cdot \mathbf{v} = 4, \quad \cos \theta = \frac{4}{\sqrt{20}\sqrt{17}}.$$

Thus the angle is approximately 77.5 degrees.



Length, distance, and unit vectors

The Euclidean norm measures length, while subtraction produces a displacement:

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}, \quad \text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

For a nonzero vector, dividing by its norm preserves direction and gives unit length:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Example:

$$\mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \|\mathbf{v}\| = 3, \quad \hat{\mathbf{v}} = \begin{bmatrix} 2/3 \\ -1/3 \\ 2/3 \end{bmatrix}.$$

The zero vector has length zero and no direction, so it cannot be normalized. Although its dot product with every vector is zero, an angle involving it is undefined.

Projection and orthogonal remainder

A projection extracts the part of a vector that points along a chosen line. Let $\mathbf{v} \neq \mathbf{0}$, and seek a parallel component $\mathbf{p} = \alpha \mathbf{v}$. We require the remainder $\mathbf{r} = \mathbf{u} - \mathbf{p}$ to be perpendicular to that line:

$$0 = (\mathbf{u} - \alpha \mathbf{v}) \cdot \mathbf{v} \implies \alpha = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}.$$

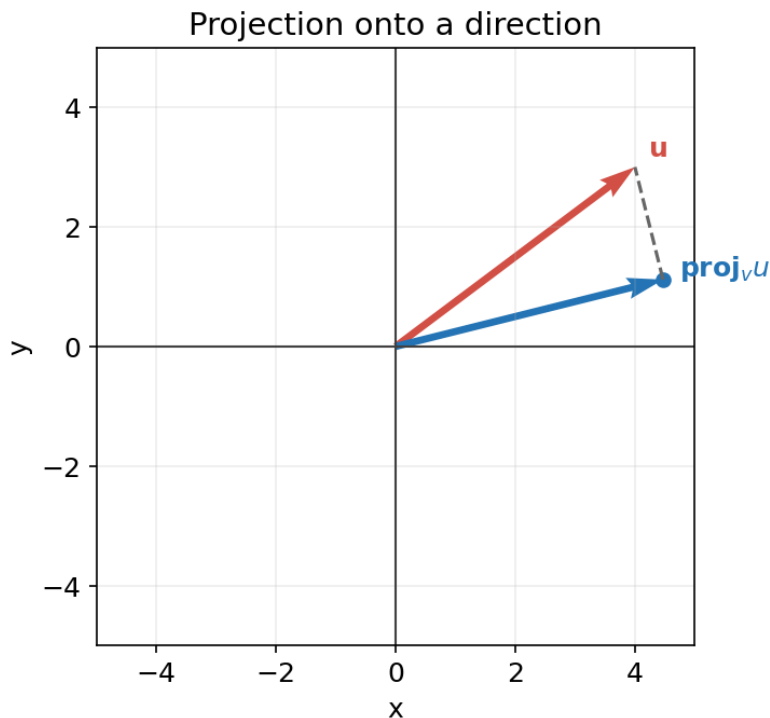
Consequently,

$$\mathbf{p} = \text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}, \quad \mathbf{u} = \mathbf{p} + \mathbf{r}, \quad \mathbf{r} \cdot \mathbf{v} = 0.$$

The plotted example:

$$\mathbf{u} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad \mathbf{p} = \frac{19}{17} \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad \mathbf{r} = \frac{1}{17} \begin{bmatrix} -8 \\ 32 \end{bmatrix}.$$

The dashed segment joins the projection to the tip of the original vector. It is perpendicular to the projection line. This is a first introduction to the geometry used later in least squares.



Cross product in three dimensions

In this course the cross product applies to two vectors in three-dimensional Euclidean space:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3.$$

Its output is another **vector**, defined by

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

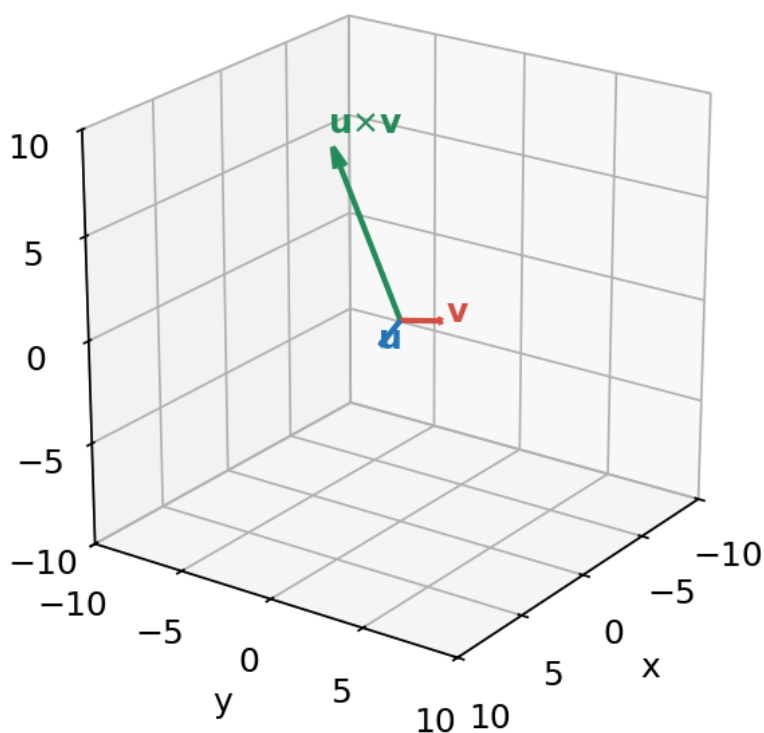
The result is perpendicular to both input vectors. For nonparallel vectors, curl the fingers of your right hand from the first vector toward the second through the smaller angle: the thumb gives the cross-product direction.

The plotted example:

$$\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \quad \mathbf{u} \times \mathbf{v} = \begin{bmatrix} 1 \\ -3 \\ 8 \end{bmatrix}.$$

Checks: $3 - 3 = 0$ and $1 - 9 + 8 = 0$, so both dot products vanish. All three plotted arrows use their actual components and the same coordinate scale.

Cross product



Cross product: area and properties

For the angle θ between nonzero vectors,

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\| \sin \theta.$$

The parallelogram has base $\|\mathbf{u}\|$ and height $\|\mathbf{v}\| \sin \theta$, so its area is the cross-product magnitude. The triangle with the same two sides has half that area.

$$S_{\text{parallelogram}} = \|\mathbf{u} \times \mathbf{v}\|, \quad S_{\text{triangle}} = \frac{1}{2} \|\mathbf{u} \times \mathbf{v}\|.$$

Swapping the order reverses direction, but leaves area unchanged:

$$\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v}), \quad \mathbf{u} \times \mathbf{u} = \mathbf{0}.$$

Parallel vectors, or any pair containing the zero vector, have zero cross product. The operation also distributes over addition:

$$\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}.$$

For $\mathbf{u} = (3, 1, 0)^T$ and $\mathbf{v} = (1, 3, 1)^T$, the parallelogram area is $\sqrt{74}$, and the triangle area is $\sqrt{74}/2$.

Matrices and the two pictures of $A\mathbf{x}$

An $m \times n$ matrix has m rows and n columns. Its entry a_{ij} lies in row i , column j . Multiplication by a column vector requires one weight for each column:

$$A \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^n, \quad A\mathbf{x} \in \mathbb{R}^m.$$

Column picture: multiply each column by its weight and add:

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

Row picture: each output component is a row dot product:

$$(A\mathbf{x})_i = \sum_{j=1}^n a_{ij}x_j.$$

For example,

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix}.$$

The column calculation is $\mathbf{a}_1 + 2\mathbf{a}_2$. The row calculation gives $1 + 4, -1 + 6, 4 + 0$. They describe the same operation.

Addition, scaling, and special matrices

Matrix addition combines corresponding entries and requires identical sizes. Scalar multiplication scales every entry:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = ca_{ij}.$$

For example,

$$2 \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 5 & 4 \end{bmatrix}.$$

The identity matrix leaves each coordinate unchanged. A diagonal matrix scales coordinates separately:

$$I_2 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ -y \end{bmatrix}.$$

The second example stretches the horizontal coordinate by two and reflects the vertical coordinate. Always check dimensions before calculating.

Reading: Strang, *Introduction to Linear Algebra*, 4th edition, Sections 1.2-1.3. Cross products supplement these sections to follow the course syllabus. General matrix multiplication and determinants come next week.

Part II: In-class problems

The theory section ends here.

Try each problem before reading its solution. A wide blank space separates the statement from the answer. Each problem starts a new page in the PDF.

Problem 1: Dot product

Let

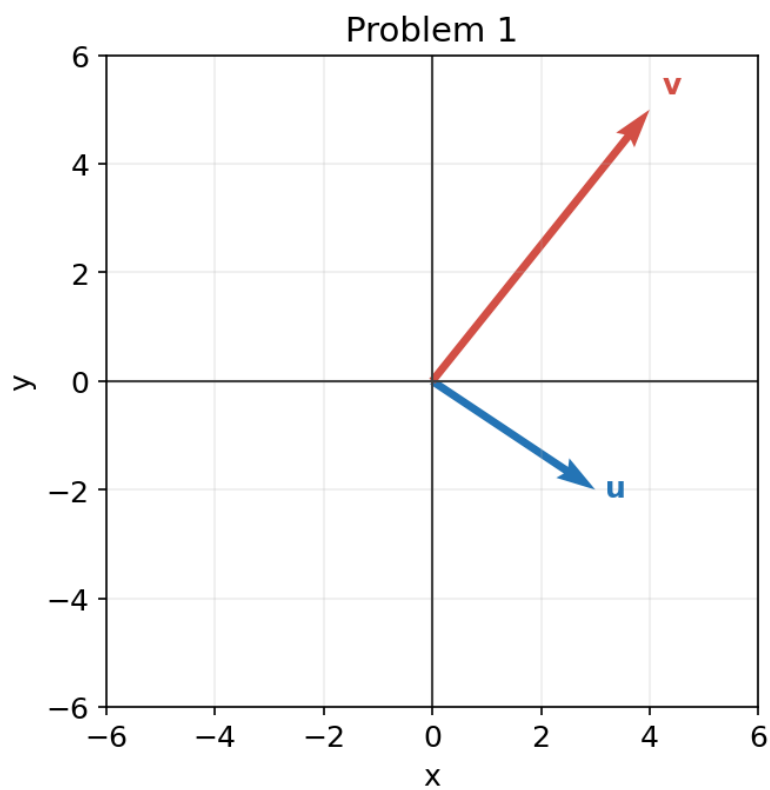
$$\mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Compute $\mathbf{u} \cdot \mathbf{v}$. Classify the angle.

Solution 1

$$\mathbf{u} \cdot \mathbf{v} = 3(4) + (-2)(5) = 2 > 0.$$

The angle is acute.



Problem 2: Length and distance

Let

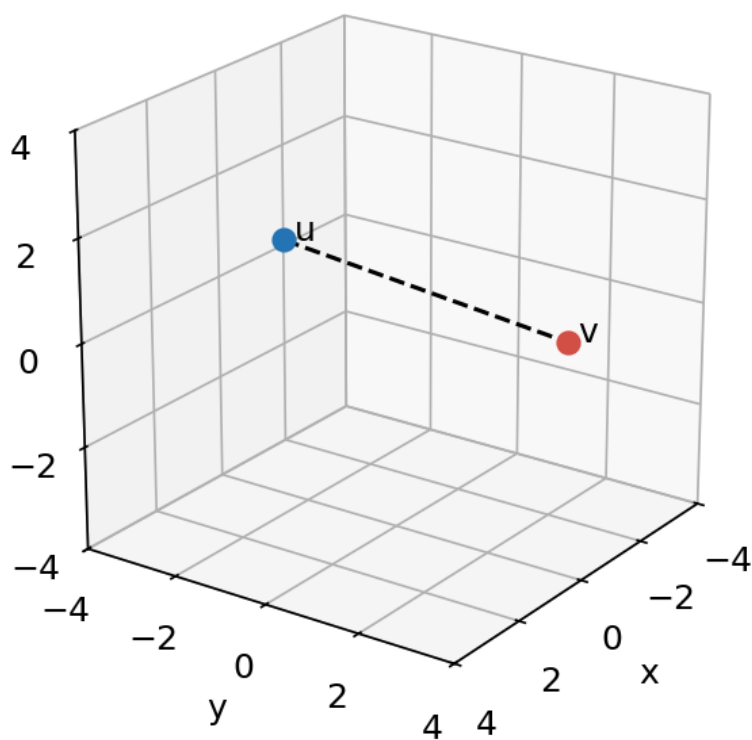
$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}.$$

Find $\|\mathbf{u}\|$ and $\text{dist}(\mathbf{u}, \mathbf{v})$.

Solution 2

$$\|\mathbf{u}\| = \sqrt{4 + 1 + 4} = 3, \quad \mathbf{u} - \mathbf{v} = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}, \quad \text{dist}(\mathbf{u}, \mathbf{v}) = \sqrt{29}.$$

Distance in $\mathbb{R}^3$



Problem 3: Orthogonality

Find k so that

$$\mathbf{u} = \begin{bmatrix} 2 \\ k \\ -1 \end{bmatrix}$$

is orthogonal to

$$\mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}.$$

Solution 3

Orthogonality requires

$$0 = \mathbf{u} \cdot \mathbf{v} = 6 + k - 4 = k + 2,$$

so $k = -2$.

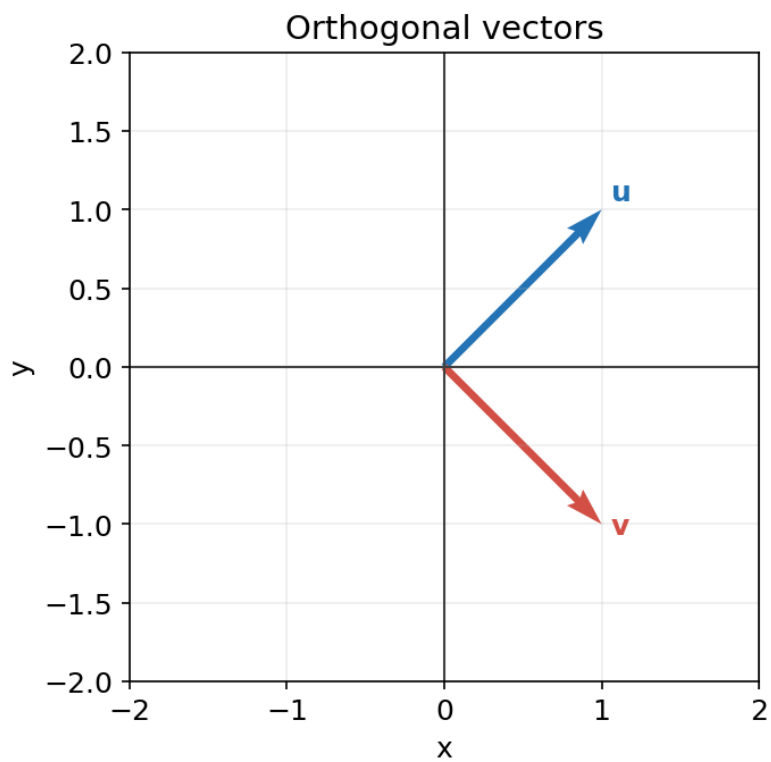
Problem 4: Angle

Find the angle between

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Solution 4

$\mathbf{u} \cdot \mathbf{v} = 0$, so $\theta = 90^\circ$.



Problem 5: Projection

Project

$$\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

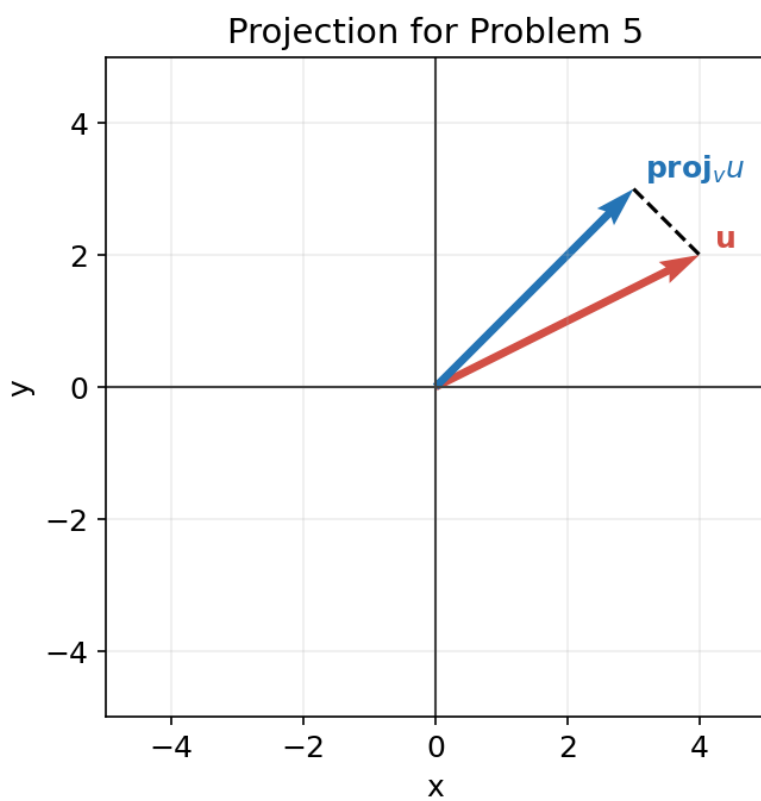
onto

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Solution 5

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{6}{2} \mathbf{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

The remainder $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is orthogonal to $\mathbf{v}$.



Problem 6: Cross product

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}.$$

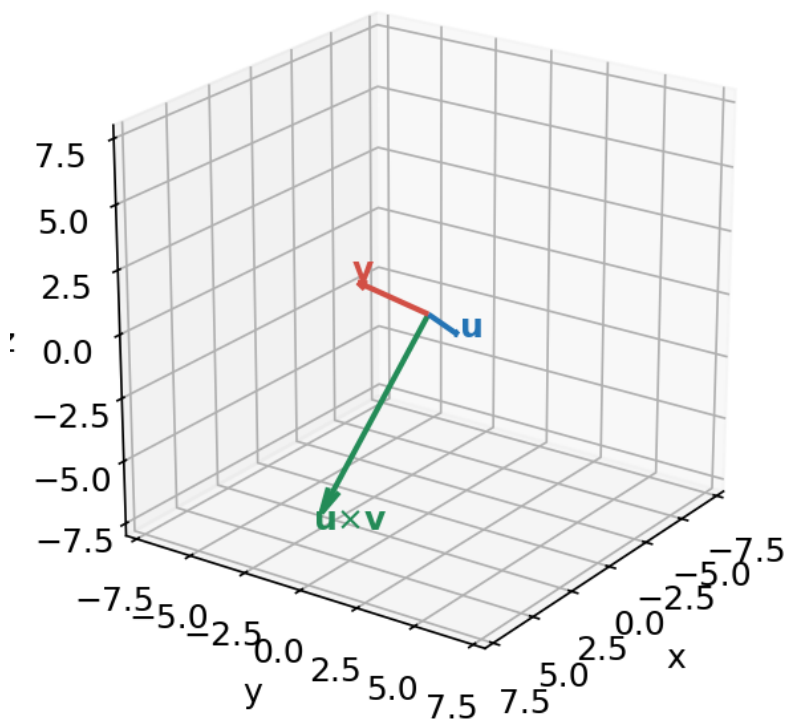
Compute $\mathbf{u} \times \mathbf{v}$.

Solution 6

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 4 \\ -2 \\ -7 \end{bmatrix}.$$

Its dot product with each original vector equals zero.

Problem 6



Problem 7: Parallelogram area

The adjacent sides are

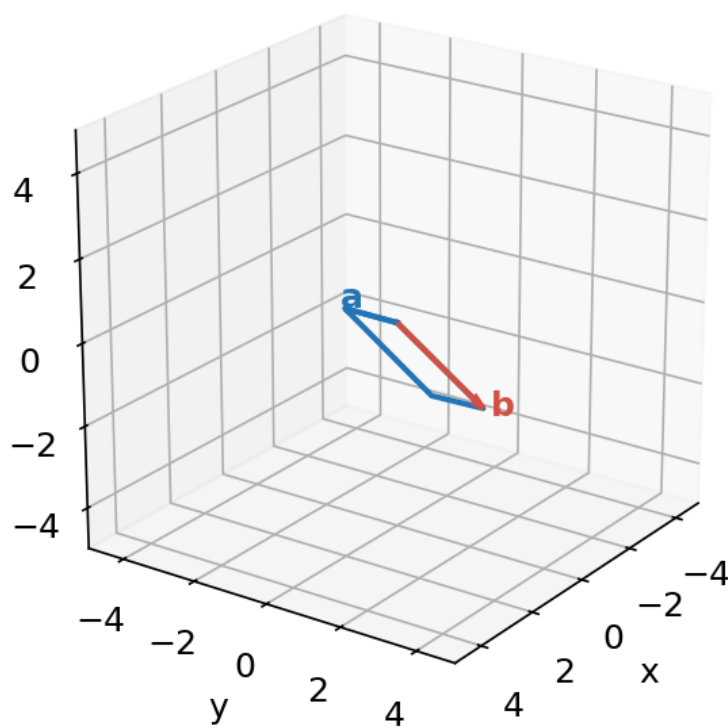
$$\mathbf{a} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}.$$

Find the area.

Solution 7

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}, \quad \text{Area} = \sqrt{54} = 3\sqrt{6}.$$

Parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$



Problem 8: Matrix size and entries

For

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix},$$

state its size, find a_{13} and a_{22} , and write columns $\mathbf{a}_1, \mathbf{a}_3$.

Solution 8

A is 2×3 , $a_{13} = 4$, $a_{22} = 3$, and

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Problem 9: Matrix addition

Let

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}.$$

Find $2A - B$.

Solution 9

$$2A - B = \begin{bmatrix} -2 & -5 \\ 7 & -5 \end{bmatrix}.$$

Problem 10: Compatibility

A is 3×2 , B is 2×3 , $\mathbf{x} \in \mathbb{R}^2$, and $\mathbf{y} \in \mathbb{R}^3$. Which of $A\mathbf{x}$, $A\mathbf{y}$, $B\mathbf{x}$, $B\mathbf{y}$ are defined?

Solution 10

$A\mathbf{x} \in \mathbb{R}^3$ and $B\mathbf{y} \in \mathbb{R}^2$ are defined. The other two products are undefined.

Problem 11: Identity and diagonal matrices

Let $D = \text{diag}(2, -1, 3)$ and

$$\mathbf{x} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}.$$

Find $D\mathbf{x}$ and $I_3\mathbf{x}$.

Solution 11

$$D\mathbf{x} = \begin{bmatrix} 8 \\ -5 \\ -6 \end{bmatrix}, \quad I_3\mathbf{x} = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}.$$

Problem 12: Compute $A\mathbf{x}$ in two ways

Let

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 4 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Use both the column picture and row dot products.

Solution 12

Column picture:

$$A\mathbf{x} = 2 \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}.$$

The row dot products give the same vector.

Problem 13: Challenge: unknown vector

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}.$$

Find $\mathbf{x}$.

Solution 13

The equations are $x_1 + 2x_2 = 7$ and $3x_1 - x_2 = 5$. Hence

$$\mathbf{x} = \begin{bmatrix} 17/7 \\ 16/7 \end{bmatrix}.$$

Problem 14: Challenge: geometry and algebra

Nonzero vectors $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are orthogonal. What follows about $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$?

Solution 14

$$0 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2.$$

Therefore $\|\mathbf{u}\| = \|\mathbf{v}\|$.